



BENCH PROTOTYPE KIT

6-aircraft synthetic feed

4 SDRs on powered hub · Laptop host

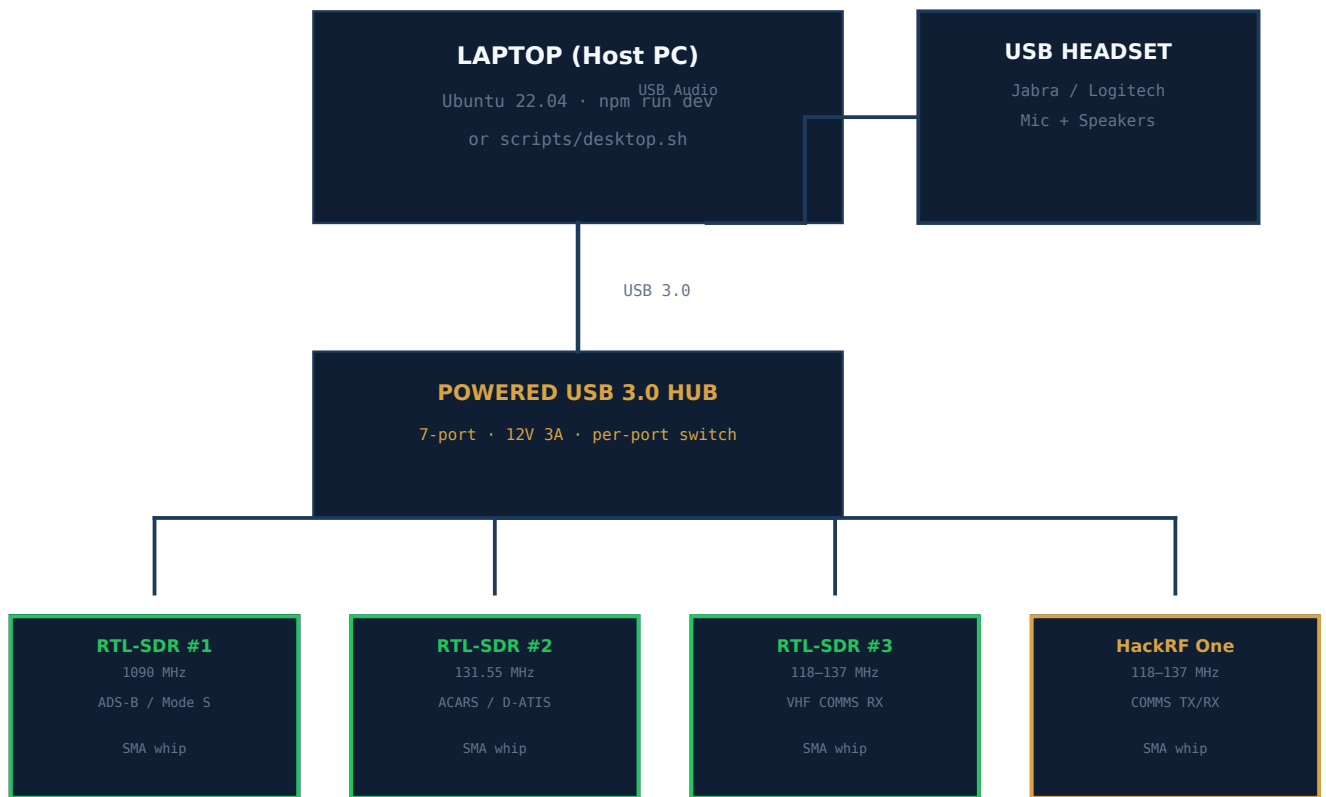
Item	Qty	Unit	Total	Notes / Source
RTL-SDR Blog V4 (1090 MHz ADS-B)	1	\$35	\$35	H.S. Electronics, Ikeja
RTL-SDR Blog V4 (131.55 MHz ACARS)	1	\$35	\$35	H.S. Electronics
RTL-SDR Blog V4 (118-137 MHz VHF)	1	\$35	\$35	H.S. Electronics
HackRF One (TX/RX 1 MHz-6 GHz)	1	\$180	\$180	Mouser / AliExpress Great Scott
Powered USB 3.0 Hub 7-port 12V/3A	1	\$22	\$22	Orico / Ugreen — Katanga
USB Headset (Jabra Evolve2 / H390)	1	\$80	\$80	Jumia / Slot
SMA male-SMA male cables 30 cm	3	\$3	\$9	Katanga cable stalls
SMA-N pigtails (future outdoor use)	3	\$5	\$15	Katanga
TOTAL BENCH PROTOTYPE		\$409	≈ ₦630k @ ₦1,550/\$ · Cash ~₦500k	

Notes: 2 dongles will run but 3 is recommended for full 3-band demo. Outdoor antennas, LMR-400, mast, industrial PC, UPS and enclosure are NOT required for bench testing – simulator provides synthetic traffic. Headset shows as standard USB audio device; no driver needed.



Bench Setup — Connection Schematic

All devices on a single powered hub · Laptop host · Indoor whip antennas supplied with dongles



POWER: Hub must be 12V powered — passive hubs brown out with 3 dongles.

Software auto-detects via `sdr_registry.rs` — chips flip Standby → Active. No code change.

What you can test with zero hardware (today): `npm run dev` → <http://localhost:1420/> (synthetic 6-aircraft feed)